L&T EDUTECH · RE PROJECT · FOLDER A

REVERSE ENGINEERING REPORT

Inside a Smartphone: Galaxy A7 (2016)

A complete teardown of model SM-A710F, from disassembly and identification to a 3D exploded view, dimensioned drawing, component analysis and Tinkercad / CAD models.



21 PARTS IDENTIFIED	**7.3 mm** BODY THICKNESS
3300 mAh LI-ION BATTERY	**8 steps** DISASSEMBLY

BATCH

28

GROUP MEMBERS

- Mohd Rehan Hasan
- Kanika Sahlot
- Ayush Kumar
- Pratishtha Dhar
- Yashveer Sharma

GITHUB LINK

github.com/RehanBuildsWay/Group-Project-PartB

Folder “RE Project”: this report, STL models and images.

0 Project at a Glance

Key facts, team and contents

21 MAJOR PARTS	**~20** PH000 SCREWS	**2** PCBS (MAIN + SUB)	**12.7 Wh** BATTERY ENERGY
151.5 mm LENGTH	**74.1 mm** WIDTH	**7.3 mm** THICKNESS	**172 g** MASS

Team: Batch 28

1	**Mohd Rehan Hasan**
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🔗	GitHub repository · Tinkercad design

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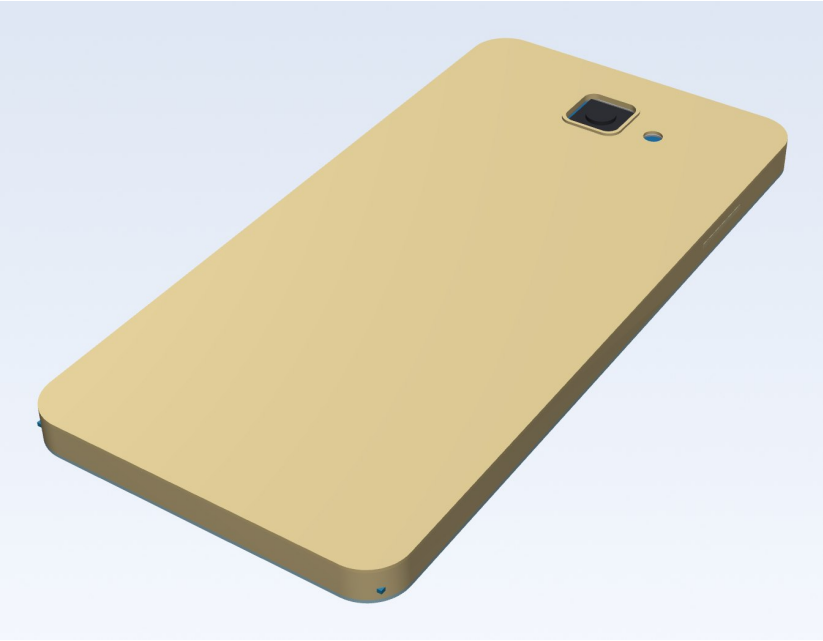


Figure 0.1 Our assembled 3D CAD model of the Galaxy A7 (2016), back view.

1 Introduction & Methodology

What reverse engineering is and how we approached it

What is reverse engineering?

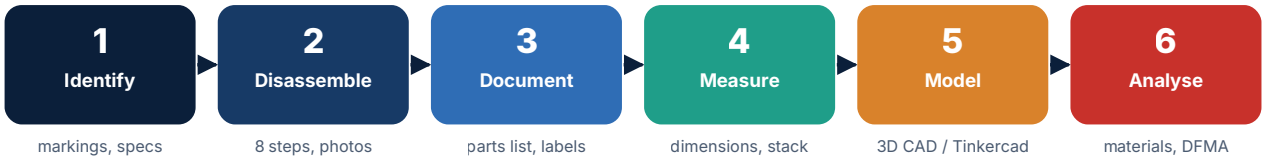
Reverse engineering (RE) means studying an existing product by taking it apart to understand **what it is made of, how the parts fit together and how it works**. The output is documentation, such as a parts list, exploded view, drawings and CAD models, that describes the product without access to the original design files.

Engineers use RE for benchmarking competitors, repair and spare parts, failure analysis, design improvement and learning. A smartphone is an ideal subject because it packs mechanical, electrical, electronic, RF and material-science design into a 7.3 mm body.

Objectives

- Safely disassemble a smartphone and photograph each stage.
- Identify the exact model and its major internal components.
- Produce a **3D exploded view**, **dimensioned drawing** and **parts list**.
- Explain how the key components work (AMOLED, SoC, battery, camera, NFC...).
- Build **3D models** in **Tinkercad** and as STL CAD files.
- Assess materials, manufacturing and repairability.

Our 6-stage method



How this document was prepared

The device was disassembled by our group, and **all teardown photographs are our own**. The college lab was then closed for construction work, so on the instructor's advice the documentation (3D models, renders, drawing and write-up) was prepared **with AI assistance**. Overall dimensions and specifications come from the manufacturer and published sources (see References). **Internal part sizes marked * are estimates** from the photographs, not caliper measurements.

2 Product Identification & Specifications

Identifying the exact model from markings found inside the phone

How the model was identified

During the teardown we read these markings:

- **“A7 REV0.4A”**: printed on the NFC / MST antenna coil.
- **“A7_DS”**: on the antenna flex cable (DS = Dual SIM).
- **“SM-A710F”**: on the PCB silkscreen and the IMEI label.
- Gold glass back, single rear camera with LED flash below it, physical home key: the 2016 A-series design.

Model **SM-A710F** = **Samsung Galaxy A7 (2016)**, international dual-SIM version.



Figure 2.1 The phone as received (back view, cracked back glass). Group photograph.

PARAMETER	SPECIFICATION (MANUFACTURER / PUBLISHED)
Model / codename	Samsung Galaxy A7 (2016), **SM-A710F** (announced Dec 2015)
Dimensions / mass	**151.5 × 74.1 × 7.3 mm**, 172 g
Build	Gorilla Glass 4 front and back, aluminium frame
Display	5.5" **Super AMOLED**, 1080 × 1920 px (~401 ppi)
Chipset (SoC)	Samsung **Exynos 7580** Octa (28 nm), 8 × Cortex-A53 @ 1.6 GHz, Mali-T720MP2 GPU
Memory	3 GB RAM, 16 GB eMMC, microSDXC (hybrid slot)
Cameras	Rear 13 MP f/1.9, AF, OIS, LED flash; front 5 MP f/1.9
Battery	Li-ion **3300 mAh**, non-removable, fast charging
Connectivity	Wi-Fi a/b/g/n dual-band, Bluetooth 4.1, NFC, GPS/GLONASS, micro-USB 2.0, 3.5 mm jack
Sensors	Fingerprint (home key), accelerometer, proximity, compass
OS (launch)	Android 5.1.1 → upgradable to 7.0

3 Tools, Safety & Disassembly Procedure

How the phone was opened

Tools used

Heat gun / hot plate	Softens back-glass adhesive (~80 °C)
Suction cup + plastic picks	Lift glass, cut adhesive
Phillips PH000 screwdriver	~20 mid-frame screws
Plastic spudger	Release board-to-board (B2B) connectors
Tweezers	Flex cables, coax plugs
SIM ejector pin	SIM / microSD tray
ESD mat / wrist strap	Protects ICs from static

⚠ Safety precautions

- **Li-ion battery**: never puncture or bend it, because damage can cause fire. Disconnect it first.
- Broken glass: wear gloves and eye protection.
- Heat gently; overheating damages the AMOLED panel.
- Use only plastic tools near the battery and flex cables.
- Work on an ESD-safe surface and keep screws sorted by position.

Step-by-step disassembly

1. **Remove the SIM tray** with the ejector pin.
2. **Heat the back glass** evenly, lift an edge with a suction cup and slide picks around to cut the adhesive (black residue visible in Fig. 4.2).
3. **Remove the back glass**. This exposes the **mid-frame** with the NFC/MST coil, camera window and antenna flex (A7_DS).
4. **Unscrew the mid-frame** (~20 PH000 screws) and fold it aside like a book (Fig. 4.3); the yellow side-key flex stays attached.
5. **Disconnect the battery connector first**, then the display FPC, coax antenna cable and camera connectors.
6. **Remove the battery**. It is glued; lift it carefully to reveal the copper display back-plate (Fig. 4.4).
7. **Remove the main PCB and sub-PCB** (one screw + B2B connector), then the cameras, vibration motor and loudspeaker.
8. **Leave the display assembly** bonded to the frame. We did not separate it, to avoid breaking the panel.

4 Teardown Photographs (Group's Own)

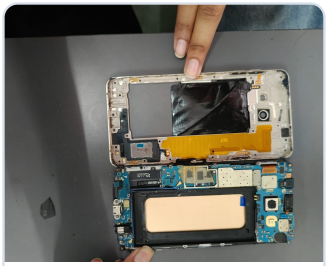
Each stage of the disassembly, annotated



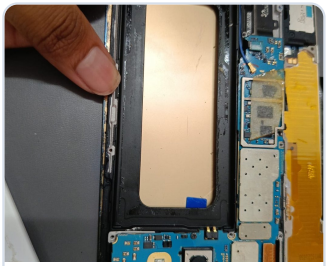
As received
Cracked gold back glass



Back glass removed
Mid-frame, NFC coil visible



Mid-frame opened
Folded aside like a book



Battery removed
PCB layout exposed

Figure 4.1 Disassembly sequence: our four key stages.



Figure 4.2 Back glass removed. The mid-frame shows the NFC/MST coil (“A7 REV0.4A”), the camera/flash windows and the antenna flex. The IMEI label is blurred for privacy.

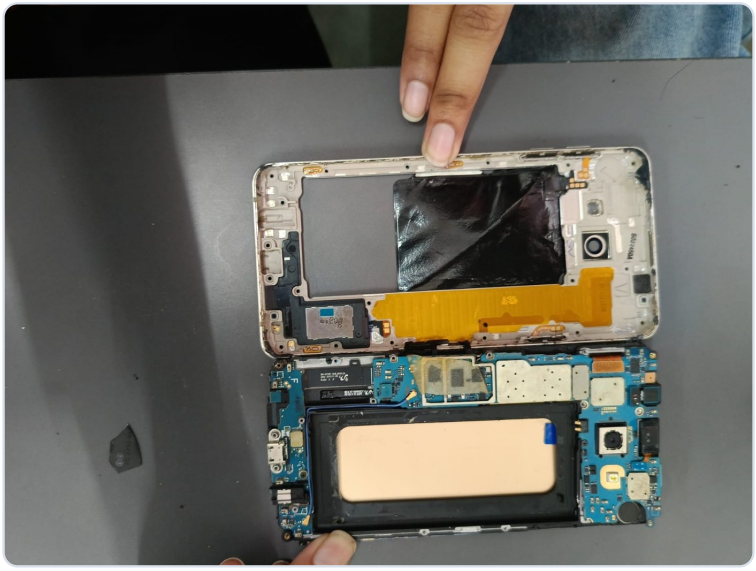


Figure 4.3 Mid-frame unscrewed and folded open (top), revealing the PCB and battery bay (bottom).

Observations

- The back glass is held only by **adhesive**; there are no clips.
- The **mid-frame** carries the NFC coil, camera window and a large **flex PCB** for the keys and antennas.
- ICs sit under **metal shield cans** (EMI shielding).
- **Coaxial cables** carry RF to antennas at both ends.
- **Kapton tape** insulates some RF chips.

4 Teardown Photographs: Internal Layout

Main PCB, sub-PCB and battery bay



Figure 4.4 Internal layout with the battery removed. The phone uses a **"U-shaped" PCB layout**: the main board runs along the top and one side, a separate sub-PCB sits at the bottom, and the battery fills the middle.

5 3D Exploded View

All major parts separated along the assembly axis

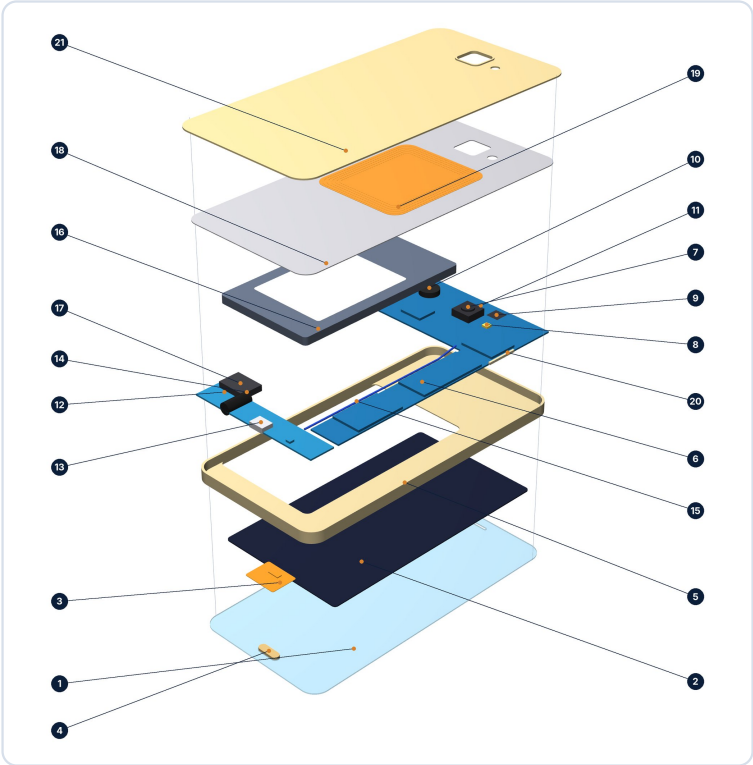


Figure 5.1 Exploded view from our 3D CAD model, from front (bottom) to back (top): glass → display → frame → PCBs & modules → battery → mid-frame & coil → back glass.

#	PART
1	Front cover glass
2	Display panel
3	Display FPC + driver IC
4	Home key
5	Frame + display bracket
6	Main PCB (logic board)
7	Rear camera
8	LED flash
9	Front camera
10	Vibration motor
11	Earpiece + proximity
12	Sub-PCB (USB board)
13	micro-USB 2.0 port
14	3.5 mm audio jack
15	Coaxial antenna cable
16	Battery
17	Loudspeaker
18	Mid-frame
19	NFC / MST coil
20	SIM / microSD tray
21	Back cover glass

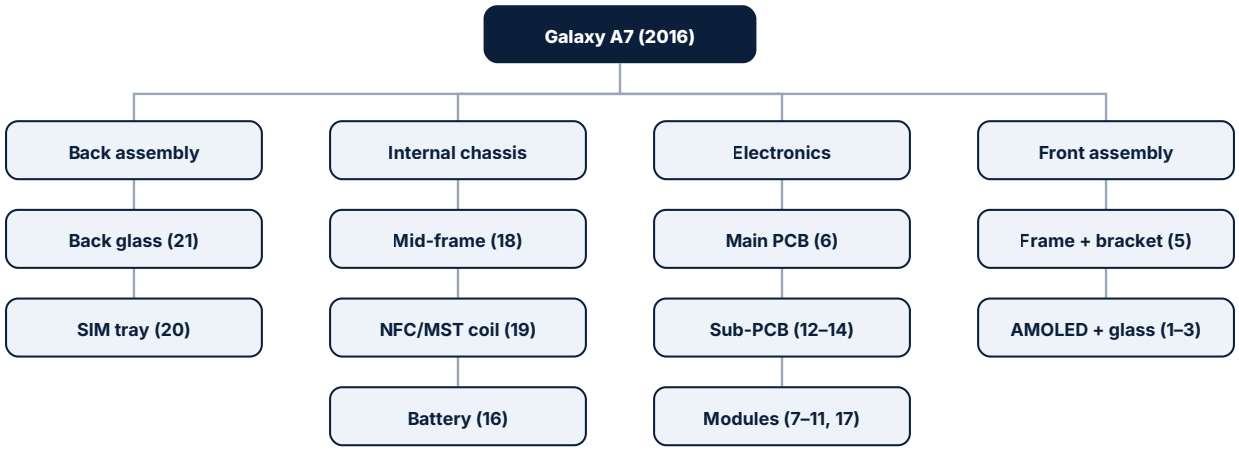
6 Bill of Materials / Component List

Every identified part with material, size and function

#	COMPONENT	TYPE / MATERIAL	SIZE (MM)	FUNCTION
1	Front cover glass	Corning Gorilla Glass 4	151.5 × 74.1 × 0.7	Protects display; touch surface
2	Display panel	5.5" Super AMOLED, 1080 × 1920, ~401 ppi	≈ 70.5 × 126 × 0.6*	Shows image; integrated touch layer
3	Display FPC + driver IC	Polyimide flex circuit	≈ 18 × 14*	Carries video signals from main PCB
4	Home key	Capacitive fingerprint sensor	≈ 12 × 4.2*	Navigation + biometric unlock
5	Frame + display bracket	Aluminium alloy (anodised)	151.5 × 74.1 × 5.9	Structural chassis; antenna segments
6	Main PCB (logic board)	Multilayer FR-4, EMI shield cans	≈ 71 × 40 + 16.5 × 91*	Exynos 7580 SoC, 3 GB RAM, 16 GB eMMC, PMIC, RF
7	Rear camera	13 MP, f/1.9, AF, OIS	≈ 10.5 × 10.5 × 4.5*	Main photo/video capture
8	LED flash	White LED	≈ 3.2 × 3.2*	Illumination for rear camera
9	Front camera	5 MP, f/1.9	≈ 6.5 × 6.5 × 2.4*	Selfie / video calls
10	Vibration motor	Coin-type ERM motor	Ø 10 × 2.7*	Haptic feedback
11	Earpiece + proximity	Dynamic receiver; IR sensor	≈ 11 × 3.6*	Call audio; screen-off near ear
12	Sub-PCB (USB board)	FR-4 with mic	≈ 71 × 13*	Charging/data interface, microphone
13	micro-USB 2.0 port	SMD receptacle	≈ 8 × 6 × 2.8*	Charging and data
14	3.5 mm audio jack	TRRS socket	Ø 5.8 × 13*	Wired headphones
15	Coaxial antenna cable	Micro-coax with U.FL plugs	Ø 0.9 × ~95*	Links main PCB RF to lower antenna
16	Battery	Li-ion 3300 mAh, 3.85 V (EB-BA710ABE)	≈ 90 × 52 × 3.9*	Energy storage ≈ 12.7 Wh
17	Loudspeaker	Micro-speaker in enclosure	≈ 15 × 11 × 3*	Ringtone / media audio
18	Mid-frame	Steel/aluminium sheet	≈ 148.5 × 71.1 × 0.4*	Holds PCB and coil; rigidity
19	NFC / MST coil	Copper coil on ferrite sheet	≈ 48 × 56*	NFC payments / Samsung Pay (MST)
20	SIM / microSD tray	Hybrid nano-SIM tray	≈ 16 × 4*	Network SIM + storage card
21	Back cover glass	Corning Gorilla Glass 4 (gold)	151.5 × 74.1 × 0.7	Rear protection; glued with adhesive

* Estimated from teardown photographs (not measured). Values without * are manufacturer specifications.

Product breakdown structure



Product breakdown structure — numbers refer to the parts list (Section 6)

7 Dimensioned Drawing

General arrangement: front, side, back and bottom views

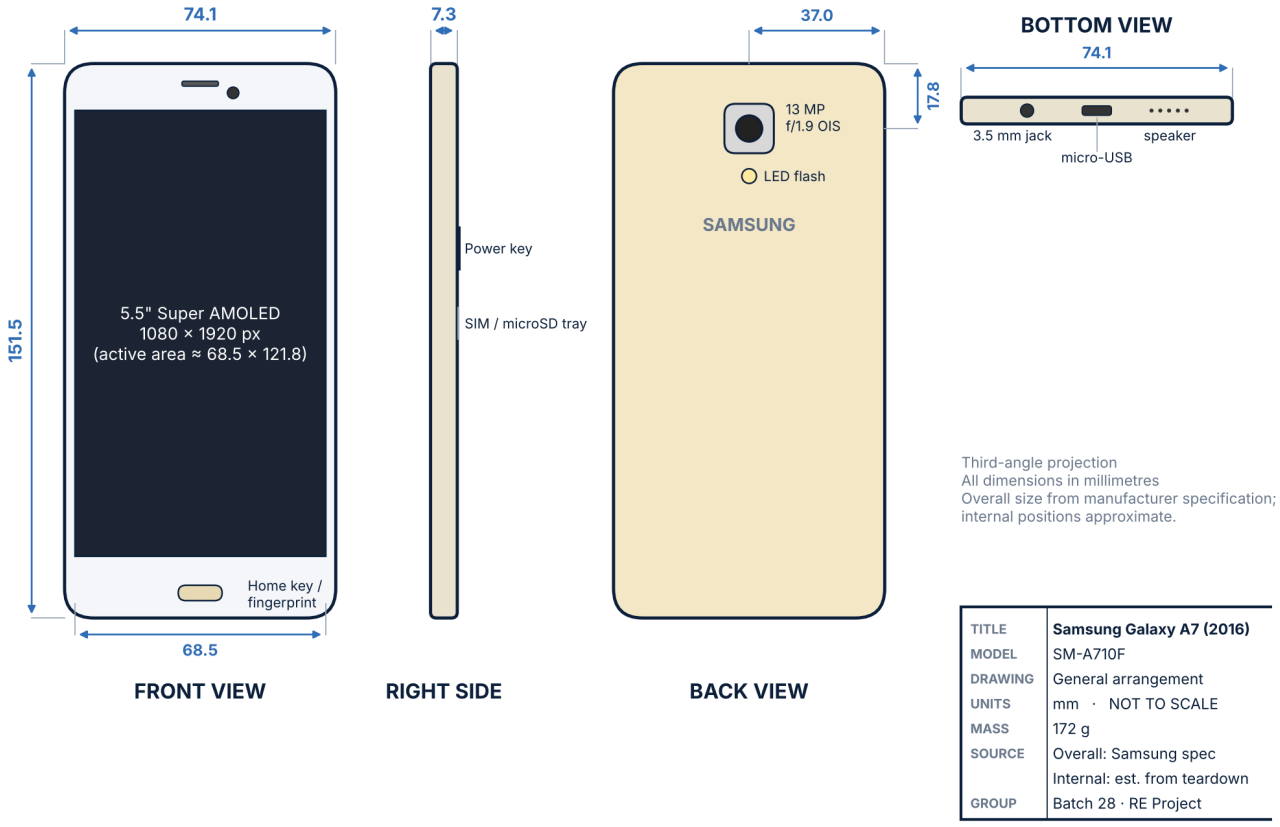


Figure 7.1 Orthographic drawing (third-angle projection), dimensions in mm. Overall size 151.5 × 74.1 × 7.3 mm (manufacturer); feature positions approximate.

Key dimensions

Length × width × thickness	**151.5 × 74.1 × 7.3 mm**
Display diagonal / active area	**5.5″ / 68.5 × 121.8 mm**
Screen-to-body ratio	**≈ 74 %**
Body volume	**≈ 82 cm³**
Mass / mean density	**172 g / ≈ 2.1 g·cm⁻³**

Worked calculations

```
16:9 screen, diagonal 5.5″ = 139.7 mm
w = 139.7 × 9/√337 = 68.5 mm
h = 139.7 × 16/√337 = 121.8 mm
Screen/body = 68.5×121.8 / (74.1×151.5)
= 74 %
Pixel pitch = 68.5 / 1080 = 63 µm (≈ 401 ppi)
```

8 Thickness Stack-up (Cross-Section)

How 7.3 mm is shared between the layers

Thickness stack-up through the battery area (vertical scale exaggerated)

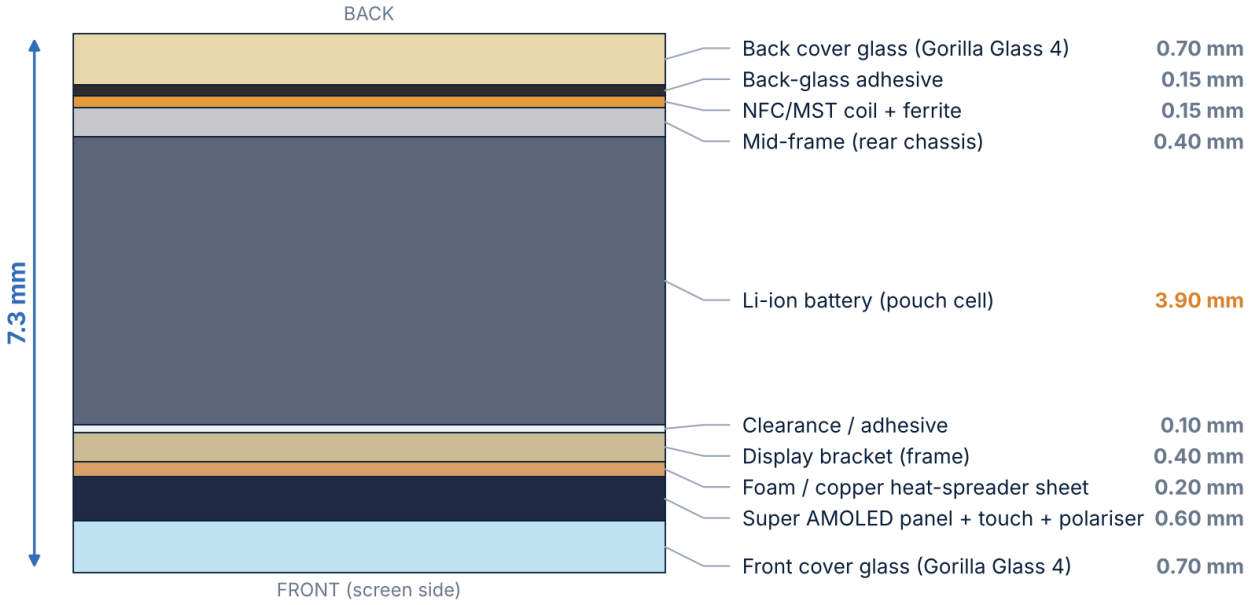


Figure 8.1 Estimated layer thicknesses through the battery area. They add up to the manufacturer's 7.3 mm total.

Observations

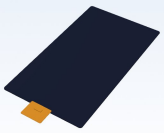
- The **battery takes over half the thickness** (3.9 of 7.3 mm). Battery capacity and phone thinness are the main design trade-off.
- The two glass covers add 1.4 mm, the price of the premium glass-sandwich design.
- A copper/graphite sheet behind the display spreads heat from the SoC.
- Clearances are only ~0.1 mm, so manufacturing tolerances must be very tight.

Engineering lesson

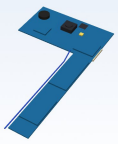
Every 0.1 mm matters in a phone. Designers choose **thin components** (on-cell touch, coin motor, pouch battery) and **stack them, not side by side**, so the PCB sits **beside** the battery (the U-shaped layout in Fig. 4.4) and doesn't add thickness.

9 Component Gallery (3D CAD)

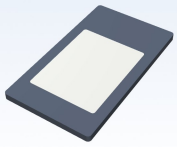
Individual parts modelled from the teardown



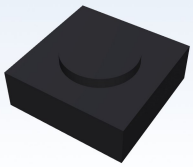
Super AMOLED panel
5.5″ · 1080×1920



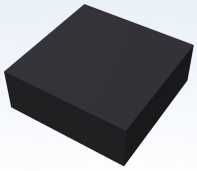
Main PCB + modules
Exynos 7580 · 3 GB · 16 GB



Li-ion battery
3300 mAh · 3.85 V



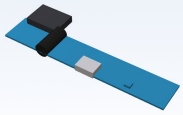
Rear camera
13 MP · f/1.9 · OIS



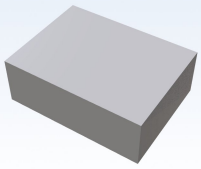
Front camera
5 MP · f/1.9



Vibration motor
Coin ERM · Ø10 mm



Sub-PCB assembly
USB · jack · speaker



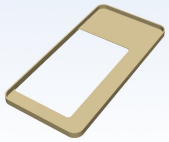
micro-USB 2.0 port
Charge + data



3.5 mm audio jack
TRRS socket



NFC / MST coil
on mid-frame



Aluminium frame
Structure + antenna



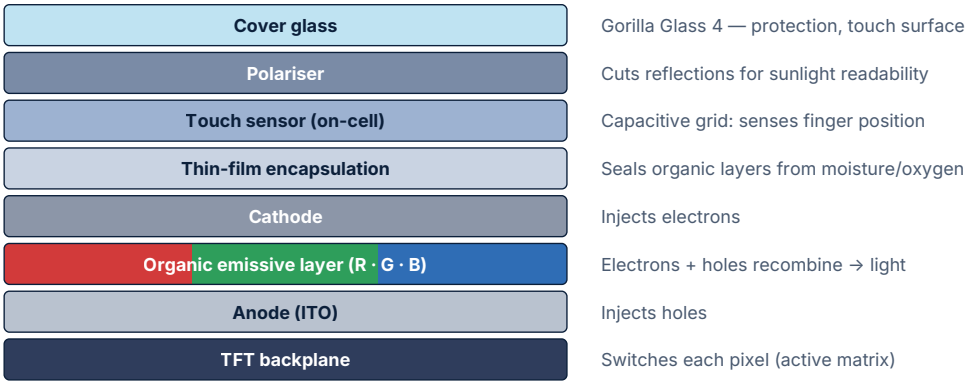
Back cover glass
Gorilla Glass 4

Simplified solid models; colours are for identification. All 21 parts are available as STL files in the GitHub “RE Project” folder.

10 Key Components Explained

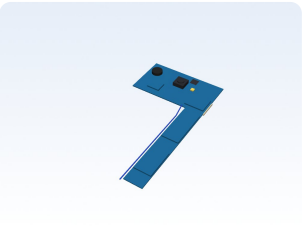
The technology behind the important parts

Super AMOLED display: how it works



Simplified AMOLED layer stack (not to scale). Each sub-pixel emits its own light — no backlight needed.

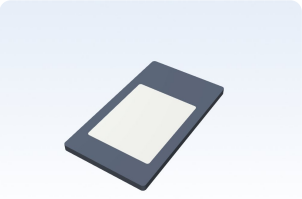
AMOLED = Active-Matrix Organic Light-Emitting Diode. When current flows, electrons (from the cathode) and holes (from the anode) meet in the organic layer and release energy as light. The TFT backplane switches each of the ~6.2 million sub-pixels. Black pixels are fully off, giving true black, high contrast and power saving.



Main PCB & SoC (Exynos 7580)

The **System-on-Chip** combines the CPU (8 × ARM Cortex-A53 @ 1.6 GHz), GPU (Mali-T720), image processor and modem on one 28 nm die. 3 GB RAM and 16 GB eMMC sit beside it under **EMI shield cans**. The **PMIC** turns battery voltage into many low-voltage rails.

multilayer HDI PCB **shield cans** **B2B connectors**



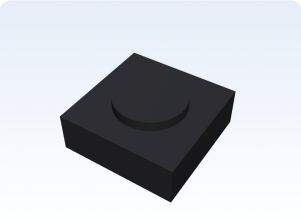
Li-ion battery (3300 mAh)

A **lithium-ion polymer pouch cell**: Li^+ ions shuttle between a graphite anode and a metal-oxide cathode. Nominal **3.85 V**, energy E = V × Q = 3.85 × 3.3 = **12.7 Wh**. A protection circuit (PCM) prevents over-charge, over-discharge and short-circuit.

non-removable, glued **EB-BA710ABE**

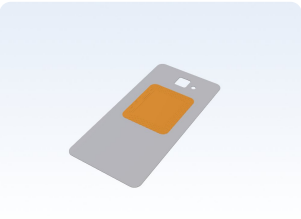
10 Key Components Explained (cont.)

Camera, NFC, motor and frame



Rear camera (13 MP, f/1.9, OIS)

A CMOS sensor converts light to charge in ~13 million pixels. A **voice-coil motor** moves the lens for autofocus. **Optical Image Stabilisation** uses gyroscope data to move the lens and cancel hand shake, which gives sharper low-light photos.



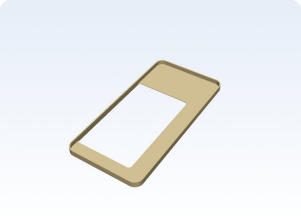
NFC / MST antenna coil

A flat copper coil on a **ferrite sheet**. **NFC** (13.56 MHz) uses magnetic induction between coils a few cm apart for contactless payments and pairing. **MST** imitates a card's magnetic stripe for Samsung Pay. The ferrite stops the field from being absorbed by the metal and battery behind it.



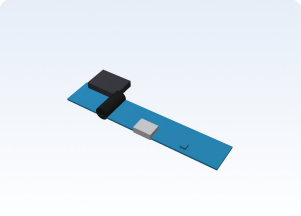
Vibration motor (coin ERM)

An **Eccentric Rotating Mass** motor: a tiny DC motor spins an off-centre weight, and the unbalanced force makes the phone vibrate. The coin shape keeps it only ~2.7 mm thick.



Aluminium frame

The metal band gives strength and also **acts as the antenna**: plastic breaks split it into separate radiating segments for cellular, Wi-Fi and GPS. The display bracket supports the AMOLED panel.



Sub-PCB (USB board)

Holds the **micro-USB port**, **3.5 mm jack**, microphone and loudspeaker contacts. Putting the high-wear USB port on a separate small board makes it **cheap to replace** without changing the main PCB.

11 System Block Diagram

How the components connect electrically

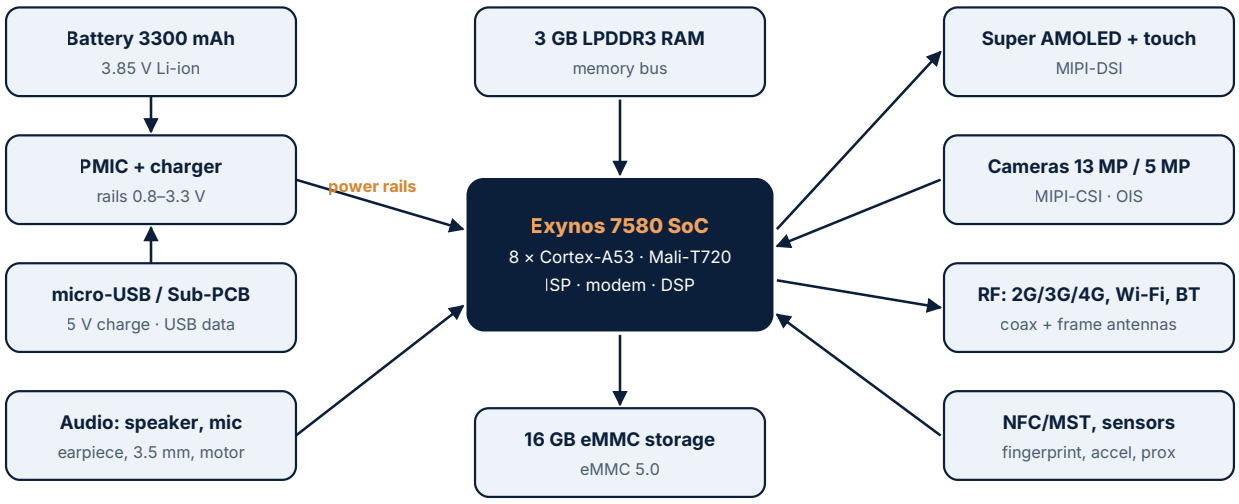


Figure 11.1 Simplified electrical architecture of the Galaxy A7 (2016).

Power path

- 5 V from the charger enters at the **micro-USB port** on the sub-PCB.
- The **charger IC / PMIC** charges the 3.85 V battery (constant current, then constant voltage).
- Buck converters and LDOs in the PMIC create the low voltages for the SoC, RAM, display and sensors.

Signal path

- The **SoC** runs Android and talks to RAM and eMMC.
- The display uses **MIPI-DSI**; the cameras use **MIPI-CSI**.
- RF goes through **coaxial cables** to the frame antennas.
- Sensors and the fingerprint reader use low-speed buses (I²C / SPI).

12 Materials & Volume Analysis

What the phone is made of and how its space is used

MATERIAL	WHERE USED	WHY (ENGINEERING REASON)
Gorilla Glass 4 (aluminosilicate)	Front and back covers	Chemically strengthened by ion exchange; scratch and drop resistant; premium feel
Aluminium alloy	Frame / side band	High strength-to-weight; machinable; doubles as antenna
FR-4 multilayer PCB	Main and sub PCBs	Glass-epoxy laminate; many copper layers for dense routing
Polyimide (Kapton)	Flex cables, insulating tape	Flexible, heat-resistant, very thin
Tin-plated steel / nickel-silver	Shield cans	EMI shielding; solderable
Ferrite sheet	Behind NFC coil	Guides magnetic flux; shields from metal
Pressure-sensitive adhesive	Back glass, battery, display	Thin bonding with some water resistance; no clips needed

Internal volume allocation (computed from our CAD model)

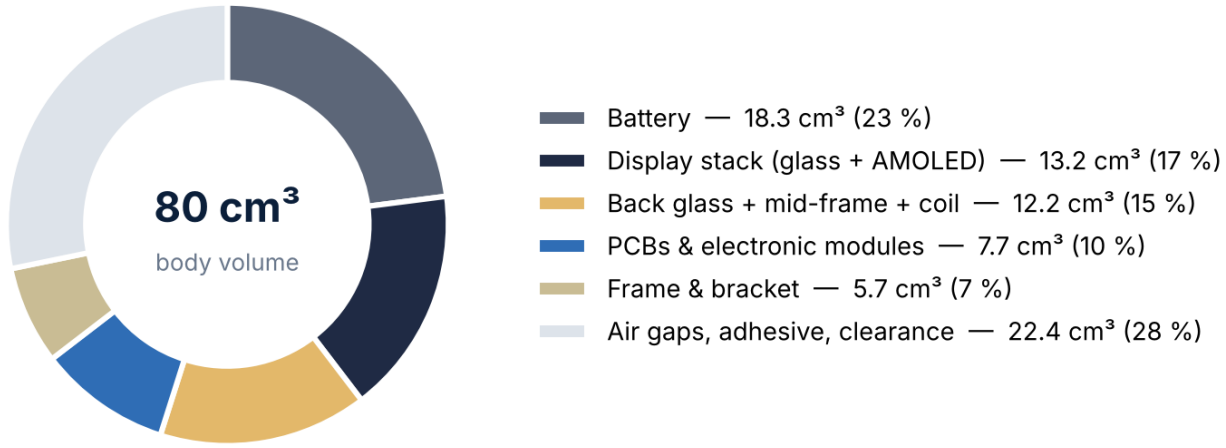


Figure 12.1 Volume shares computed from our simplified CAD model. The battery is the single largest internal part.

13 Repairability & Design for Manufacture

Engineering assessment from the teardown

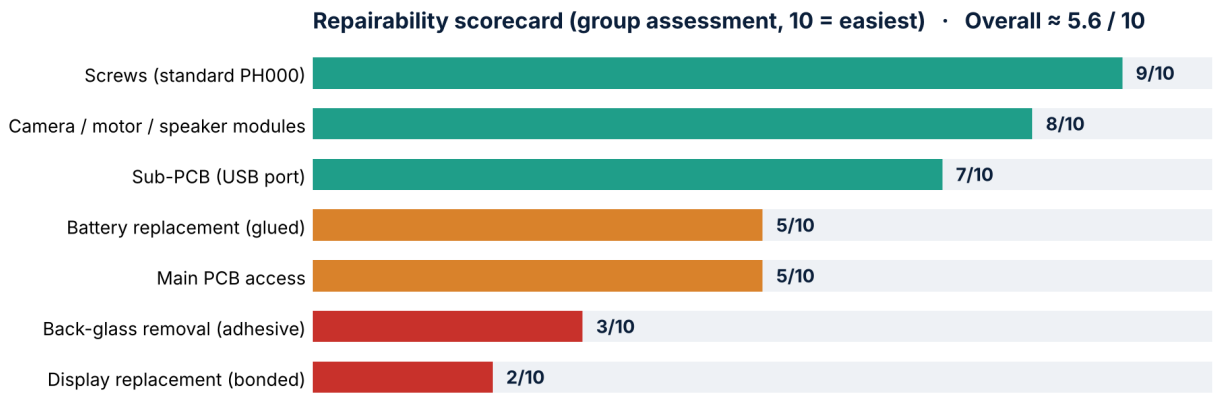


Figure 13.1 Our repairability assessment of each disassembly task.

Design for manufacture (DFM) observations

- **SMT assembly**: tiny 0201/0402 parts placed by machine.
- **Board-to-board connectors** let modules be fitted quickly on the line.
- **Standardised screws** (one size, PH000) speed up assembly.
- **Modular sub-PCB** isolates the high-wear USB port.
- Revision labels (A7 REV0.4A, A7_DS) track parts by model.

Suggested design improvements

- Pull-tab adhesive strips for easier battery removal.
- USB-C instead of micro-USB (reversible, faster charging).
- Clips plus a thin gasket instead of full adhesive on the back glass.
- Label connectors on the PCB to guide repair.

14 Tinkercad & 3D CAD Models

Our 3D design and downloadable STL models

TINKERCAD DESIGN LINK

https://www.tinkercad.com/things/bef88fv2Qjg/editel?returnTo=%2Fdashboard&sharecode=X_HNz-SgkPMsiebUFrg5TXTvmoao9KiFXkagSA6_ggs

Tap the link or scan the QR code to open our group's 3D design in Tinkercad (sign-in may be needed).

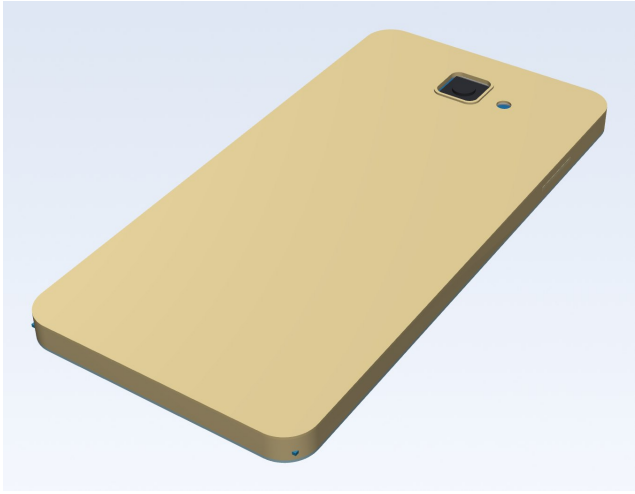


Figure 14.1 CAD model, back.

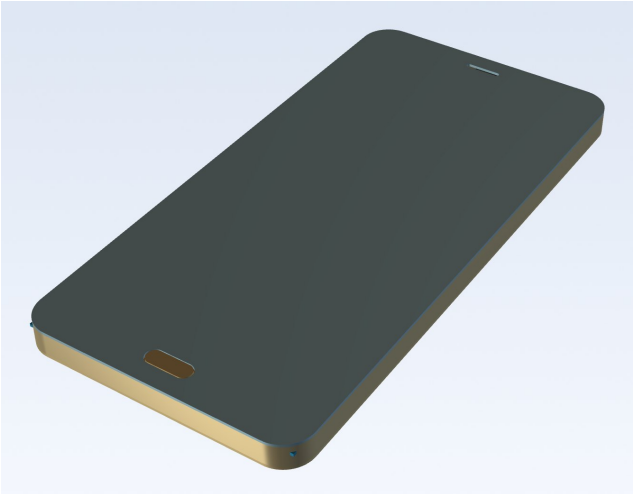


Figure 14.2 CAD model, front.

STL files in the GitHub “RE Project” folder

`00_Full_Assembly_Galaxy_A7_2016.stl`	Complete phone
`00_Exploded_View_Galaxy_A7_2016.stl`	Exploded layout
`01 … 21_*.stl`	One file per part

Built to the manufacturer's overall dimensions; internal parts are simplified blocks sized from the teardown photos.

Import into Tinkercad

1. tinkercad.com → **Create → 3D Design**
2. **Import** (top right) → choose any **.stl** file
3. Units: **millimetres**, scale 100 %
4. Colour parts and use **Align / Group** to build an exploded view.

STL files also open in Fusion 360, FreeCAD and SolidWorks, and in GitHub's built-in 3D viewer.

15 Conclusion & Learning Outcomes

What the reverse-engineering exercise taught us

- From internal markings (A7 REV0.4A, A7_DS, SM-A710F), the phone was identified as a **Samsung Galaxy A7 (2016), SM-A710F**.
- It was disassembled in 8 steps; **21 major parts** were identified, photographed and documented.
- We produced a **3D exploded view**, a **dimensioned drawing** (151.5 × 74.1 × 7.3 mm), a **thickness stack-up**, a **volume analysis**, a **Tinkercad design** and **STL CAD models**.
- We studied the key technologies: **Super AMOLED**, **Exynos 7580 SoC**, **Li-ion battery** (12.7 Wh), **OIS camera**, **NFC/MST**, **ERM motor**.
- Main design trade-offs: glued glass and battery give a thin premium body but reduce repairability (≈ 5.6/10); the metal frame doubles as an antenna; the battery dominates thickness and volume.

Learning outcomes

- Safe disassembly practice (ESD, Li-ion handling)
- Identifying products from part markings
- Exploded views and orthographic drawings
- 3D modelling in Tinkercad / CAD
- Linking materials and manufacturing to design decisions
- Teamwork and technical documentation

References

1. GSMArena: Samsung Galaxy A7 (2016) full specifications. gsmarena.com/samsung_galaxy_a7_(2016)-7759.php
2. Wikipedia: Samsung Galaxy A7 (2016).
3. VRM24: How to disassemble Samsung Galaxy A7 2016 SM-A710.
4. iFixit: Samsung Galaxy A7 (2016) device page.
5. Group teardown photographs, Batch 28.

Batch 28 · Mohd Rehan Hasan · Kanika Sahlot · Ayush Kumar · Pratishtha Dhar · Yashveer Sharma
github.com/RehanBuildsWay/Group-Project-PartB